LIFELINE — Iteration 1 approach note

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1. From story to numbers

All 50 cases are synthetic, generated from deterministic random streams keyed by `(group_seed=18, case_no, module, attempt)`. Cases 00–19 are tier S, 20–39 M, 40–45 L, and 46–49 X. Except for the specified traps, tier X uses L ranges. We choose sizes independently within the course's ranges, then enforce scenario-specific shapes and quotas. Crates weigh 1–20 kg and have stored values 10–100; capacity is 10–70% of their total weight. **There is no day multiplier or storm multiplier in the workbook.** A road step from u to v costs `minutes(u,v) + penalty(v)`; closures exclude roads instead of changing costs.

Study and story cue	Numerical construction and rationale
1. Delta Flood: “flat delta”, “roads go under water”, “food and water matter most”	Set average degree to **2.2–3.2**, with at least **45%** of locations assigned penalty >= 2 and **3–5** disjoint closure waves. Force at least **60%** of crates into food/water. Connected but moderately sparse roads model raised crossings.
2. Mountain Earthquake: “few roads”, “injured need medicine and tools”, “bursts of urgent calls”	Average degree **2.0–2.6**; road times **5–60 min**; at least **52%** medicine/tool crates. Sample aftershock-clustered arrival ticks and high-weighted priorities until the mean is at least **6.0**.
3. Coastal Cyclone: “shelters”, “short and overlapping windows”, “radio full of noise”	At least **30%** of locations are shelters; **3–5** closure waves. Rescue windows are **5–60 min**, clustered around 1–3 peaks; average strict overlap >= 2. Radio uses **30%** replacement noise and includes all **38** allowed symbols.
4. Urban Flash Flood: “many alternative routes”, “calls arrive faster”, “rare equipment”	Average degree **3.5–5.0** and short roads of **1–25 min**. Force at least **15%** rare crates. On M/L, alarms arrive at **1.5–1.8 per tick**, exceeding the one-per-tick service rate.
5. Landslide Corridor: “single highway”, “collapse cuts off communities”	Build a highway containing roughly **2/3 or more** of locations with short branches, maximum degree **4**, average degree <= **2.4**, and HQ eccentricity >= floor(V/4) roads. Put a critical bridge in one of **3–5** waves, leaving at least one target unreachable.
6. Outbreak Camp: “medicine and water”, “supplies spoil”, “beacons repeat”	Force at least **62%** water/medicine crates; at least **60%** expire by hour **300**. Reuse beacon messages, padding with underscores until _ occupies at least **30%** of the radio log.

2. Planted objects and exact small-case claims

- **Roads and closures:** a spanning tree is generated first, ensuring pre-closure connectivity; extra roads establish each study's graph density. Waves are disjoint sets of valid road numbers, each with 1 to max(1, floor(E/4)) members. For study 5 a verified bridge is forced into a wave.
- **Tour:** choose distinct integer sites in the `[0,999]` grid; start a nearest-neighbour tour at site 0, breaking ties by lower site number. Apply only improving 2-opt swaps; store its complete permutation and recomputed *closed Manhattan* length. Thus L/X planted length never exceeds the nearest-neighbour baseline. This is **not** a claim of global tour optimality at large sizes.
- **Sensor vertex cover:** choose `sensor_k` cover nodes first, then draw every edge with at least one endpoint in that cover. **Safety 3-SAT:** choose a binary assignment first, then construct each three-switch clause so at least one literal is true. **Rations:** choose a nonempty subset first, then set `rations_target` to its exact sum.
- **Radio and claims:** split logs into ordered parts of <=10,000 symbols; record six patterns of lengths 2, 4, 8, 12, 20, 40 and count overlapping occurrences. Set `edit_bound` to the exact Levenshtein distance computed by DP. For every tier-S case only, exhaustive algorithms compute the six `claims` optima (cargo, window count/value, drum, closed tour, and minimum cover), independently rechecked by V5. Larger cases have no purported exact-optimum claims.
- **Variety and validation:** eight S cases are constructed to defeat ratio-greedy knapsack and eight to defeat fewest-roads cost; M/L rule/switch bins are planned as **7 low, 11 middle, 8 high**. Checks V1–V8 recompute constraints, including trap conditions. `validator/tests/` supplies valid single-case samples and deliberate single-defect mutations whose rejection is recorded in `validation_report.txt`.

3. Four hostile trap cases

Case	Hostile construction	Why it is difficult; required check
18-46, dock order	**1,500–3,000** crates ordered monotonically by value, weight, expiry and serial; value/kg is also monotone.	A naive first-pivot quicksort can degrade on ordered inputs. **V8** verifies at least three of five monotone fields and the special size range.
18-47, best-looking crate	Capacity **40**; two **20 kg, value 100** crates, plus a **1 kg, value 10** ratio-greedy decoy and poor-value fillers.	Ratio-greedy takes the decoy, then only one good crate: **110 vs optimum 200 (55%)**. **V8** recomputes greedy and exact knapsack values.
18-48, shortest looks cheapest	A long highway (**560–700** positions) forces graph distance >=500; a local gadget has a costly two-road route and cheap four-road alternative.	Fewest-road BFS differs from minimum-cost routing by >=1.5x for a target. **V8** checks the diameter lower bound and both route costs.
18-49, near miss	**50,000–100,000** characters, at least 90% covered by repetitions of the **one-symbol block** A; four patterns have near-matching prefixes at >=half of start positions.	Massive prefix-matching pressure stresses naive pattern search. The 1-character block is allowed by the spec's “at most 8” condition. **V8** verifies coverage and near-miss counts.

4. What will be hard in Iteration 2

Scenario	Two difficult relief decisions and the specific data property
1. Delta Flood	**B route planning:** multiple closure waves can force detours and change reachability. **A inventory:** heavy, differently valued food/water crates make density-greedy unreliable under integer capacity.
2. Mountain Earthquake	**B route planning:** sparse 2.0–2.6-degree roads provide few alternate paths. **D emergency queue:** aftershock bursts plus high priorities demand stable urgent-first release rather than simple FIFO.
3. Coastal Cyclone	**C rescue scheduling:** strongly overlapping short windows require weighted interval DP rather than highest-value greedy. **E radio logs:** 30% character noise and many symbols create difficult phrase-search/edit-distance workloads.
4. Urban Flash Flood	**D emergency queue:** 1.5–1.8 arrivals per tick exceed service capacity, so the backlog grows. **F network planning:** dense alternative roads require choosing minimum *cable* spanning connections, not fastest routes.
5. Landslide Corridor	**B route planning:** a wave can disconnect a target completely; an algorithm must return “no route.” **F network planning:** the severed corridor changes which sites are reachable, so cost-minimal cable must be computed on the actual active component.
6. Outbreak Camp	**H hard planning:** planted but non-obvious safety/ration feasibility challenges brute force as sizes rise. **E radio logs:** repeated beacon symbols generate many overlapping phrase matches and strongly skewed Huffman frequencies.

Implementation boundary: This note documents **Iteration 1 data and proposed Iteration 2 decisions**, not completed implementations of modules A–I. All three members must verify the final descriptions and be able to explain the full system in the viva.